Quality Assurance Plan

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1 Quality Policy Statement

To consistently meet and strive to exceed the agreed requirements of our clients in the most cost effective manner by providing a high standard of Management, Technical expertise, Workmanship and timely completion of the project.

This will be achieved by employing highly professional people with drive and determination to work as a team with our Clients, Architects, Consultants Contractors and Suppliers in a collaborative and non-adversarial manner.

This is comprehensive Quality Assurance Plan. Individual checklist will be applicable based on scope of supply and installation of various items listed in this document.

2 Plan Preparation Procedure

An established system of Management of Quality shall be followed on the project, aiming to achieve and maintain quality throughout the tenure of the project. This shall be achieved through the following major steps:

- Establish a clear and project-specific standard of procurement and work practices.
- Resolve specific problem areas covering technical, product or system issues.
- Investigate new methods of work and ideas.
- Innovation in the existing procedures and practices.
- Continuous Improvement.

There shall be proper planning, monitoring, control and co-ordination of the following achieves this:

- Design Intent.
- Documentation.
- Contractor Selection.
- Contract.
- Project Program / Schedule.
- Procurement Strategies.
- Construction and Review of Methodologies.

2.1 *Design*

- Assess all designs with respect to any impediments that may be potentially experienced during execution for delivering quality works on the project, keeping in mind the skill levels, materials and products commonly available in the market. These impediments could range from overall build ability issues to specific material selections.
- As far as possible, the use of exotic materials shall be avoided to ensure that the local skill levels available in the market are able to deliver a good quality job for the entire project. Any such exotic material, if necessary for the project, shall be treated as a separate area for monitoring on quality, with training compulsorily required from the supplier / applicator.

2.2 *Documentation*

- All pertinent incoming / outgoing documents for the project will be controlled in accordance with the DOCUMENT CONTROL PROCEDURE, which will ensure their effective recording and review. It will also provide for the subsequent controlled distribution of Documents and superseding / destroying / removal of obsolete documents.
- The Document control procedure will identify all requirements for long term storage, maintenance and retrieval of archive records including defect liability period [DLP].
- Ensuring the adoption of a change management system to capture all changes to the project during PMC tenure, including changes on account of quality requirements.
- Ensuring that the Contractors submit their scope-specific Quality Plan for the project, including a detailed list of Quality documents planned to be maintained for the project. Review and approve their Quality Plan after aligning it with the quality requirements specified in their contracts and any other special project requirements.
- Ensuring that the contractors implement and safely maintain all their quality documents and records as per document control procedure and submit to PMC / Consultants for inspection and approval as and when required.
- Ensuring that quality-related performance of contractors is suitably linked to bill payments for contractors.
- Generating hand over documentation, performance histories, audit reports, defect rectification reports and testing & commissioning reports during hand over phase.

2.3 *Contractor Selection*

- All potential Contractors for the project will be analysed in detail by means of a Pre-Qualification system before recommending for inclusion in the tender list. The Pre-Qualification system has a well laid-out system of analysing the Contractors' commitment to Quality, both by way of policy and by way of track record. Based on their Quality policy, systems and procedures, experience in similar works and experience of Client / Consultants / PMC with the Contractors in any previous projects, a joint assessment is made regarding their eligibility to participate in the bidding process for the project.

2.4 *Contracts*

- Quality requirements for the project will be included in the tender / bid documents stage itself so that the bidding Contractors are made aware of the expectations from the initial stages of the bidding process and price for the same.
- At each subsequent stage of the bidding process up to the award of contract, quality requirements will be discussed and reviewed with the bidding Contractors to ensure clarity.
- Level of Quality supervision to be deployed on the project as a minimum, shall form part of the Contract documentation.
- All the guarantees, warranties and performance parameter set out in the specifications / agreed with the clients and consultants form part of Contracts to ensure that the set out standards are met.

2.5 *Program / Schedule*

- A detailed contract program shall be obtained from the Contractors and reviewed to ensure that adequate time & resources have been allocated to enable works to happen as per the quality requirements for the project set out in the approved Quality Plan.
- Quality deviations, their reasons and remedial measures shall form part of program / schedule reviews.
- Photographic reports of quality deviations and their rectifications shall be used as the main tool to monitor and ensure closure of such instances.
- Reasons for the quality deviations in the activities shall be analysed and documented; the team shall agree the mitigation strategy for the same.

2.6 *Procurement Strategy*

- Goods and services will be procured from acceptable Contractors / Agencies, shortlisted through the Pre-Qualification system and finally selected through a detailed bidding process, with each step giving separate focus to quality parameters of Contractor evaluation.
- A thorough Technical Bid Analysis [TBA] would be carried out to ensure all the specifications are well understood by the Contractors before they are awarded the subcontract
- The appointed Contractors / Suppliers will be required to provide appropriate Quality documents with the means to confirm the Quality is being competently addressed and implemented at all levels.
- Necessary inspection / test schedules for bought-out equipment / specialized items by Client / Consultants / PMC at the suppliers or Contractor's premises shall form part of the Contractor's quality documentation.
- As far as possible, the use of exotic materials shall be avoided to ensure that the local skill levels available in the market are able to deliver a good quality job for the entire project. Any such exotic material, if necessary for the project, shall be treated as a separate area for monitoring on quality, with training compulsorily required from the supplier / applicator.

2.7 *Construction*

Methods of monitoring the implementation of Quality Plan by Contractors will include, but will not be limited to, the following:

- Auditing Contractor's Quality Management Systems and their implementation. Involving Contractors quality department and senior management personnel.
- Submission of detailed project-specific work method statements (WMS) by Contractors for each major activity and for any critical activity (major or minor).
- Review and approval of work method statements (WMS) and monitoring their implementation through audits and site inspection sequences.

- Witnessing activities, inspections and tests.
- Surveillance and site inspection.
- Checklists.
- Mock up construction and Approval.
- Material Inspection and Testing Reports [MITR].
- Site Instructions [SI].
- Control of Non-conforming Products.
- Weekly review of Quality deviations, action plans thereof and completion of the same.
- Monthly update and review with Contractor senior management regarding quality system strength and any outstanding quality issues.
- Review of records and Data.
- Closeout Procedures.

2.7.1 Auditing & implementation of PMC and contractor's quality systems

PMC Project Manager will be in-charge of arranging audits of all activities to ensure compliance with the operating procedures. He will ensure the conduct of these periodic audits in accordance with the checklists / records / Material Test Reports as set out in this document. The audit program will cover the following categories of audits.

- ✓ Internal Audits of PMC Site.
- ✓ External Audits of suppliers and Contractors where appropriate by Project Manager.

All quality audits whether internal or external, will be carried out in accordance with the quality system auditing procedures. These procedures will also cover the distribution or reports and method of follow up and close out of findings.

A summary of visits and audit and comments of auditing personnel shall be issued monthly, attached as part of Monthly report. Findings and discussions of Monthly Status Review Meeting shall be implemented and used for closeout and regularly issued in the next Monthly Status Review Meeting.

2.7.2 Reviewing, approval & implementation of work method statements (WMS)

Contractors will be instructed to submit a WMS prior to the commencement of any works in which they shall describe in detail, each step that will be taken to execute the works along with the list of tests, checklists, etc. that would be used to monitor those works and the quality parameters to be achieved. These method statements will be reviewed and approved before allowing works to proceed. The approval in normal course shall be provided by PMC. However, in the case of any special works needing inputs from experts, the Architects / Consultants / Clients review shall be sought before giving approval to the Contractors.

2.7.3 Witnessing activities, inspections and tests

There shall be a list of tests prepared and submitted by Contractors, which need witnessing as per their Contracts. This list shall be reviewed by PMC with Consultants and Client to include any tests missed out or additionally required, before it is finally approved to the Contractors.

Contractor shall give sufficient intimation for witnessing inspection and testing and will provide the details of the tests need to be conducted and observations to be made. However PMC will provide sufficient guidelines for ensuring proper Inspection and tests and will also have an overall supervision of works to ensure the Quality of works.

Inspection and testing of various materials shall be done at specified intervals, viz., daily, weekly, stage-wise, etc. basis the guidelines set out in this manual.

2.7.4 Surveillance and site inspection

PMC engineers will implement the checklists, tests and other tools in the Quality Management System to ensure adequate surveillance and monitoring of Contractors works and ensure that Contractors are performing to the specified standards.

2.7.5 Checklists

PMC will bring to the project its array of standard quality checklists which cater to almost all regular works and practices in the industry for civil, mechanical and electrical works. Further customized checklists / fresh checklists will be prepared where needed, based on the project-specific design, specifications and Contractor skill levels.

2.7.6 Mock up construction and approval

PMC will arrange for the Contractors to submit samples of materials and arrange to construct mock-ups as per specifications, prior to start of any major works. These shall be set up for inspection and approval by Client / Consultant to ensure complete alignment with their expectations.

2.7.7 Site instructions [SI]

In case of the works getting executed such that it does not conform to the specifications and / or the method statements and results in faulty works/unacceptable quality, PMC will issue Site Instructions to the Contractor asking for the rectification of works with reference to the specifications and methods agreed. In the event of product failure even after following all agreed methods, the issue shall be investigated in detail with the help of the Consultants, Client and Contractors' technical teams to carry out a root cause analysis and implement remedial measures.
Site Instructions shall be tracked for closure and the status of closure shall form part of the internal project review of PMC.

2.7.8 Control of non-conforming products

In case of major deviations, repeat quality deviations (small or big), a Non-Conformance Report shall be issued to the Contractor and the defective works shall be quarantined and clearly identified. The Contractor shall submit a formal action plan to rectify the works and / or strengthen his quality control system to avoid recurrence.

2.7.9 Review of records and data

The Records and Data generated in the due course of project will be reviewed regularly for compliance and areas of improvement.

2.7.10 Close out procedures

Project Close-Out Plan shall include review of compliance record of quality parameters by the Contractors, including close-out of all NCRs, availability of all checklists, material test certificates, etc.

- ✓ Technical Closeout: This shall be done package-wise and all quality documentation as per package contract shall be collated, reviewed and filed. This file shall form part of the basis for acceptance of contractors' works and issuing Practical Completion Certification to them. Contractual close-out: Ensuring obligations in contracts are closed out and all their accounts settled based on their compliance with quality requirements and closure of all quality NCRs, punch lists, snag lists, etc.
- ✓ Establishment of Hand over procedure for Contractors to PMC and PMC to Clients. These procedures to include quality parameters as per package contractors.
- ✓ Handing over to the Operation and Maintenance Team with list of do's and don'ts for maintaining quality of the building and facilities.

The above shall be achieved by making these points part of the project finalization meeting so that all parties concerned have equal understanding of expectations.

2.7.11 Defects liability period

PMC are not part of the DLP management process for contractors. However, PMC shall do the following as a support for the Client's facility management team to manage the contractors' DLP as per the conditions of contract…
PMC shall provide a list of all vendors on the project, with name of contact person, addresses, telephone numbers, email ids and any other form of communications available with the vendor for establishing the long term service set up and contact during the DLP.

3 Quality Requirements for Trade Contractors

3.1 *Contractors Quality Plan*

The following points are minimum procedures the contractors are expected to comply with,

- Use of the Contractors own checklists previously approved and PMC checklist spot-checked.
- Visual inspection.
- The Contractor shall deliver as part of his submission a project specific Project Quality Plan (PQP) which will be reviewed during the selection process.
- The PQP shall cover the works of the Contractor plus any of his Contractor's petty contractors.
- PMC's representatives and / or other members of the consultant and Client team will spot-check the works and sign the relevant records to confirm what has been spot-checked, however the responsibility for ensuring the works are to specification will remain with the contractor. Usually for any special / critical new work, an approval is sought by PMC from consultants to check quality and compliance with design intent – this will become part of quality audit.
- Should major or repeat non-conformity be identified by PMC a Corrective Action Request [CAR] will be issued. The Contractor will be required to close out the CAR within the prescribed period.
- Should the Contractor fail to effectively implement the agreed actions, then PMC will take whatever action is necessary to regain control of the works and/or standard of workmanship of the product all at the contractor's expense.

4 Checklist

Following checklists shall be prepared based on the project progress and the requirements. As the project progresses the following list shall be revised as and when required by the project team in concept with the Project Manager.

4.1 *Civil Works*

- ✓ Surveying and setting out
- ✓ Excavation
- ✓ Back filling
- ✓ Soling
- ✓ Anti termite
- ✓ PCC and Plum Concrete
- ✓ Reinforcement
- ✓ Formwork
- ✓ Concreting
- ✓ Post pour
- ✓ Water Proofing
- ✓ Blockwork / brickwork
- ✓ Internal / external plastering
- ✓ Painting
- ✓ Flooring works
- ✓ Tiling and Dado
- ✓ Counters
- ✓ Door & windows
- ✓ Railing Works
- ✓ Ironmongery
- ✓ Any other as may be defined by the Project Manager during execution

4.2 *Electrical Works*

- ✓ HT Panel
- ✓ Transformers
- ✓ LT Panels including power control centres & motor control centres
- ✓ Lighting and power distribution boards
- ✓ HT and LT cables and conductors
- ✓ HT and LT Switchgear
- ✓ Lighting point wiring
- ✓ Light fittings / luminaires
- ✓ Power points
- ✓ Conduiting – Floor
- ✓ Conduiting
- ✓ Earth conductor
- ✓ Lightning arrestor system
- ✓ Cable trays
- ✓ Earthing
- ✓ Raceways
- ✓ Bus ducts
- ✓ Uninterrupted power supply systems
- ✓ DG sets with accessories
- ✓ Any other as may be defined by the project manager during execution

4.3 *Utility Equipment and Systems*

Water treatment plants, Boilers/Steam Generators, Air Compressors, Chilling plants, HVAC systems, Lifts and escalators, Wastewater (Effluent) and sewage treatment plants, EOT/OHT cranes, Process gases generation/storage systems, Cooling Towers, Pumps, Fire fighting systems.

- ✓ Manufacturing stage inspections
- ✓ Factory acceptance test
- ✓ Incoming material inspection for capital goods
- ✓ Incoming materials checks for piping, valves, fittings, instrumentation
- ✓ External Laboratories Test/Certification.
- ✓ Installation & pre-commissioning checklists
- ✓ Commissioning checklists
- ✓ Performance tests
- ✓ Utility piping, plumbing & sanitary
- ✓ Installation checks for piping, valves, fittings, instrumentation (Including Pressure test, radiography and other relevant tests as defined by the Project Manager)
- ✓ Commissioning checks for piping, valves, fittings, instrumentation
- ✓ Any other as may be defined by the Project Manager during execution.

4.4 *External Works*

- ✓ Site grading
- ✓ Roads
- ✓ Parking
- ✓ Drains
- ✓ Walkways and pathways
- ✓ Curbs
- ✓ External Signage
- ✓ Any other as may be defined by the project manager during execution

4.5 *General*

- ✓ Final Check and take-over

5 Formats

Various Checklist / Method Statement Formats

Sr. No.	Checklist
A	**Civil and Interior Work**
1.	Excavation
2.	Backfilling (Murum & Soil)
3.	Soling Work
4.	Anti-Termite Treatment
5.	PCC
6.	Plum Concrete
7.	Reinforcement- Footing, Retaining Wall , Raft
8.	Reinforcement-Slab/Beam
9.	Reinforcement – Column/Lift Shaft
10.	Form Work – Slab, Beam
11.	Form Work – Column, Lift Shaft
12.	Formwork- Footing, Retaining Wall, Raft
13.	Concrete-pour Card
14.	Post Concrete / P.C.C / R.C.C
15.	Waterproofing
16.	Masonry-Brick, Block ,AAC
17.	Internal Plaster , Gypsum Plaster, POP
18.	External Wall Plastering
19.	Concealed Drainage -Supply Pipes Chiselling Work
20.	Concealed Water -Supply Pipes Chiselling Work
21.	Fixing Doors
22.	Aluminium Windows
23.	Waterproofing Work- Toilets/Wet Areas
24.	Granolithic Flooring
25.	Platform/Counters
26.	Floor Tiling
27.	Wall Tiling
28.	Internal Wall Painting
29.	External Wall Painting
30.	Plumbing and Sanitary Fixtures
31.	Staircase
32.	Tremix Flooring
33.	Lift Installation
2	**IMF Formwork Inspection**
3	**Fabrication and Erection**

Sr. No.	Checklist
4	**External Development**
5	**Road Works**
6	**Interior**
6.1	Gypsum/ POP Punning
6.2	False Ceiling Works
6.3	S. S. Railing Work
6.4	M S Fabrication Work
6.5	Dry Cladding Work
7	**Electrical**
7.1	Switchyard
7.1.1	Installation - Lightning Arrester
7.1.2	Installation - Bus Post Insulator
7.1.3	Installation - Current Transformer
7.1.4	Installation - Potential Transformer
7.1.5	Installation - Isolator without Earth switch
7.1.6	Installation - Circuit Breaker
7.1.7	Installation - Isolator with Earth switch
7.1.8	Installation- Control & Relay Panel
7.1.9	Installation - Transformer
7.2	HV/LV Cable
7.3	D.G. Set
7.4	Bus duct Transformer & DG
7.5	Installation - Earthing Station
7.6	LT Panel
7.7	Distribution Boards
7.8	CCTV Camera Installation
7.9	G.I./M.S. Conduits
7.10	Raceways/Junction Boxes
7.11	Light/Power Point Wiring
7.12	Light Fittings
7.13	Raw Point, Light Switches/Sockets
7.14	Switchboards/MCC panels inspection & Test report
7.15	Installation - Lightning Protection System
8	**HVAC**
8.1	Air Cooled Chiller
8.2	Quality Control–Cooling Tower
8.3	Quality Control–Pump

Sr. No.	Checklist
8.4	Quality Control–Piping
8.5	Hydro Test Report
8.6	AHU - Quality Control
8.7	Ducting -Quality Control
8.8	Commissioning Record Sheet - Pumps
8.9	Commissioning Record Sheet & Test Report – Chiller
8.10	Commissioning Air Handling Units
8.11	Air Balancing Report
9	**Fire Protection System**
9.1	Quality Control Piping
9.2	Fire Hydrant Post
9.2	Fire Hose Box Installation
9.3	Fire Sprinkler Installation
9.4	Portable fire extinguisher installation
9.5	Fire fighting with grooved fittings
9.6	Fire Alarm Detector Installation
9.7	Module Installation
9.8	Response Indicator Installation
10	**Piping**
10.1	Arc Welding
10.2	Die Penetration Test
10.3	Pipe Installation
10.4	Wrapping and Coating
10.5	Holyday testing
10.6	Welding Joint Radiography Test
10.7	Hydro Test
10.8	Pipe Painting
11	**Instrument**
11.1	Impulse Piping Installation
11.2	Pneumatic Air Line Installation
11.3	Instrument Installation
11.4	Cable Laying
11.5	Cable Trays